

Manual Test Cases

- **TC_01: E2E Integration Test: Portal Entry Point**
 - **Priority:** High
 - **Preconditions:** Browser cache and cookies are cleared; no active booking session exists.
 - **Steps:**
 1. Navigate to the Homburg city portal: <https://www.homburg.de/>.
 2. In the main navigation menu, click on "Rathaus".
 3. From the dropdown list, select "Stadtverwaltung", then navigate to "Online-Dienste".
 4. Locate the graphical tile labeled "Termin mit Bürgerbüro/Führerscheinstelle vereinbaren" and click it.
 - **Expected Result:** The widget's landing page (Step 1 of 6) loads successfully in a new browser tab. The address bar displays the target URL: (<https://termine-reservieren.de/termine/homburg/>).
- **TC_02: Boundary Value Validation: Date of Birth Triad (Step 5)**
 - **Priority:** High
 - **Preconditions:** Active session; user is on Step 5. Mandatory text fields (Vorname, Nachname, E-Mail, E-Mail-Wiederholung) are populated with valid test data. Privacy consent checkbox is checked. Date of birth fields (Tag, Monat, Jahr) are initially empty.
 - **Steps:**
 1. Clear the "Jahr" field, enter an out-of-range year (e.g., 1895), and unfocus the field.
 2. Verify the status marker next to the "Jahr" field and the state of the "Reservieren" button.
 3. Clear the "Jahr" field, enter a valid birth year (e.g., 1995), and unfocus.
 4. Verify the status marker next to the "Jahr" field and the state of the "Reservieren" button.
 - **Expected Result:**
 1. **Step 1-2 (Invalid Year):** The system identifies the year as being outside the valid range. A red error marker ([!]) appears next to the "Jahr" input, form validation fails, and the "Reservieren" button becomes disabled.
 2. **Step 3-4 (Valid Year):** A green success marker ([✓]) appears next to the "Jahr" field. The interface permits submission without Day/Month values; all mandatory validations are satisfied, and the "Reservieren" button becomes active.
- **TC_03: UX Isolation & Legal Modal Logic**
 - **Priority:** Medium
 - **Preconditions:** User is on Step 1 of the booking widget. The global session timer in the footer is active and counting down.
 - **Steps:**
 1. Scroll to the bottom of the page and click the "Datenschutzerklärung" link in the footer.

2. Visually confirm the appearance of the modal window, the correctness of the German text, and the background overlay (blocking background interaction).
 3. Locate the "Hier" hyperlink within the modal text (linking to the full DSGVO document) and click it.
 4. Return to the initial widget tab and click the "Schliessen" button (or the close [X] icon in the top-right corner).
- **Expected Result:**
 1. **Step 2:** The legal information modal opens as an overlay. The background is dimmed, and interactions with the Step 1 department selection buttons are completely disabled.
 2. **Step 3:** The full privacy policy document opens in a new browser tab. The booking session and the modal window on the original tab remain active and state-preserved; the footer timer continues to count down without interruption.
 3. **Step 4:** The modal window closes smoothly. The user is returned to Step 1 in its initial state, and no previously selected parameters are reset.
 - **TC_04: Input Validation for 'Jahr' with Special Characters Acceptance**
 - **Priority:** High
 - **Preconditions:** Active session; user is on Step 5. Mandatory personal data fields are filled with valid test data. Privacy consent checkbox is checked.
 - **Steps:**
 1. Navigate to Step 5 (Personal Data).
 2. Locate the "Jahr" (Year) input field.
 3. Enter a valid year followed by a special character (e.g., "1995,").
 4. Click outside the input field to trigger the blur/validation event.
 5. Observe the validation status marker next to the "Jahr" field and the state of the "Reservieren" button.
 - **Expected Result:**
 1. The system must sanitize the input and reject any non-numeric characters.
 2. A red error marker (![!]) must appear next to the "Jahr" field.
 3. The specific error message "Bitte geben Sie ein zulässiges Geburtsdatum ein" must be displayed.
 4. The "Reservieren" button must remain disabled.
 - **TC_05: Map Interaction & External Google Maps Navigation**
 - **Priority:** Medium
 - **Preconditions:** Active booking session; user is on Step 3 (Branch Selection).
 - **Steps:**
 1. Locate a branch card and click the "Karte anzeigen" button.
 2. Confirm that the interactive Leaflet map and blue marker pin render correctly below the card.
 3. Click on the blue branch marker pin on the map.
 4. Verify that the popup info window opens with two action buttons: "Standort auswählen" and "Hierhin navigieren".
 5. Click the "Hierhin navigieren" button.

- **Expected Result:**
 1. Steps 1-2: The interactive map renders with zooming and panning enabled.
 2. Step 3-4: The info window opens smoothly over the marker pin.
 3. Step 5: Google Maps opens in a new browser tab (target="_blank") with direct route directions. The original booking tab remains active, keeping the session timer and selected options intact without page reload.